

Cross-Module Integration and Pipeline Contracts

Liu Bowen

3 Discussion

3.1 Cross-module limitations

The integration contract improves provenance but does not remove biological or technical limitations. The module scores have different estimands and native scales, so their co-location in one table is not numerical harmonization. The number of independent GSE176206 animals is small, cell counts are unequal, and unknown-animal observations are descriptive sensitivity data. Risk interpretation also depends on the released gene lists, coverage rules, reference thresholds, aggregation policy, and model version. Finally, Geneformer embedding shifts cannot yet be used as direct inputs to the expression-based scorers.

3.2 Future work: a multi-dataset integration framework

Future work should extend the contract to additional datasets and animal-condition structures through versioned intake schemas, explicit biological-unit keys, and machine-checkable provenance and quality-control fields. Cross-assay transportability should remain distinct from independent external validation. For the prediction arm, a Geneformer-derived state should be connected to the scoring axes only after a validated predicted-expression interface is available. This would permit in-silico screening of perturbation conditions followed by multi-axis state evaluation, while longitudinal functional and safety endpoints would still be required before a universal safe region could be claimed.

4 Methods

4.1 Data architecture and pipeline contracts

The integration layer was designed to preserve the native analysis contract of each module. Cells were treated as measurement units, whereas donors or animals were treated as the biological inference units. This distinction was required because arithmetic aggregation of cell-level scores and scoring of aggregated expression profiles are different estimands. Accordingly, the integration layer did not refit any Youth, Identity, or Risk model and did not transform their raw score scales.

For each module, the contract recorded the input representation, preprocessing steps, scoring unit, aggregation rule, model version, quality-control fields, and source artifact. The cell-level Youth model retained its native scoring and donor summary. The v3.1 donor-pseudobulk Youth framework retained M1 as the primary operational model and M4 as a separate high-stringency sensitivity model. The two operating points were never averaged or ensembled. The Identity and Risk modules were likewise retained as separate named

outputs.

The canonical biological-unit identifier comprised four fields:

```
dataset_id | age_group |  
exact_treatment_arm | animal_label
```

Exact treatment-arm labels were retained throughout the pipeline. Repeated numeric animal labels were not interpreted as evidence of pairing, because animal labels were nested within treatment arms in the GSE176206 application. A pairing status was recorded only when supported by an explicit sample map; otherwise the comparison was labelled unpaired or unknown.

The canonical output schema required the biological-unit identifier, dataset and condition fields, module name and version, native score name, aggregation rule, coverage and quality-control fields, evidence level, and source file. Missing module outputs were left missing and were not imputed. Unknown-animal rows were retained only as descriptive sensitivity observations and were excluded from the primary known-animal inference layer.

The Risk definition was represented by two independent axes, namely *Pluripotency Risk* and *Genomic Stress Risk*. Inflammation was not included in the formal final output. The versioned gene list, scoring scripts, and donor-summary table were registered as source artifacts; inclusion of Risk values in the primary integrated table additionally required the input-data reference, software environment, coverage and threshold rules, unknown-animal policy, and clean-checkout reproducibility record.

Geneformer was represented as a separate perturbation/state-transition prediction arm. Its embedding shift is not a predicted expression matrix and therefore cannot be passed directly to an expression-based Youth, Identity, or Risk scorer. Any future connection would require an independently validated predicted-expression adapter.